KEDAR DAMALE

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Portfolio Website · GitHub · LinkedIn

SUMMARY

Applied Data Scientist with 1.5 years of experience building end-to-end AI/ML solutions across fintech, financial automation, pharma, and medical research. Experienced in machine learning, Generative AI, multi-agent systems, and data pipeline development, with a focus on translating research and experimentation into deployable, production-grade applications.

SKILLS

- **Programming and Data** : Python, SQL, Databricks, DuckDB, NumPy, Pandas, Polars
- **Machine Learning** : Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Generative AI** : LangChain, LangGraph, Multi-Agent Systems, RAG, MCP, Human-in-the-Loop (HITL)
- **Deep Learning** : PyTorch, CNNs, YOLO, Transfer Learning, Model Training and Evaluation
- **Data Engineering and MLOps** : ETL/ELT Pipelines, MLflow, FastAPI, Docker, AWS, CI/CD, Git/GitHub
- **Mathematics and Statistics** : Probability and Statistics, Linear Algebra, Calculus,

WORK EXPERIENCE

ML and GenAI Engineer **Jan 2026 - Present**
GlobalSpace Technologies Ltd. *Navi Mumbai, MH, India*

Budget Automation and Incentive Calculator

- Replaced a per-employee XGBoost modeling approach with a unified, hierarchy-aware target allocation system designed to scale from 20–30 employees to thousands of MRs. The solution is under review by pharmaceutical companies including Himalaya and Zydus.
- Built a top-down budgeting workflow that allocates targets based on historical performance and capacity, supports time-bound revision requests, reconciles conflicting requests, and routes proposed changes for managerial approval.
- Developed a DAG-based incentive calculation engine with an AI agent that converts policy PDFs into executable logic at runtime and explains employee-level incentive outcomes, enabling business users to update incentive policies without specialist admin support.

PatGPT

- Rebuilt a pharmaceutical sales analytics platform around an agentic architecture for multi-step forecasting, simulation, prediction, and historical analysis across 3M+ sales and tender records.
- Implemented human-in-the-loop review checkpoints and evidence-grounded responses to support verifiable, multi-step commercial analysis and recommendations.
- Added streamed responses, safe retry handling, and checksum-verified data promotion and rollback. The platform is now used at Zydus.

Automation Developer **May 2025 - Jan 2026**
Propelligence Advisors Pvt. Ltd. *Thane, MH, India*

- Automated purchase-register to GSTR-2B invoice reconciliation, addressing inconsistent vendor names, invoice identifiers, and date formats that made exact-match approaches unreliable.
- Engineered a large-file reconciliation pipeline using fuzzy matching, match threshold, date/value tolerances, Polars processing, and constant-memory report generation.
- Processed 10,000+ invoices in 12 seconds versus 5–6 days manually, reducing processing time by over 99.9%. Achieved 100% precision with no false positives till this date. Integrated the module into an audit suite adopted by two CA firms and approximately 20 auditors.

PROJECTS

1. Chessablanka: Chess Position Analyzer **GitHub Link**

- Built a computer-vision tool that converts chessboard photos into playable positions, eliminating manual board transcription for chess analysis and college competition preparation.
- Trained a custom YOLOv8 model using an augmented dataset of 1,700+ images to detect chess pieces and reconstruct board positions from photos.
- Achieved 98.57% detection accuracy across 500+ test images and integrated Stockfish for position evaluation and best-move recommendations; used by the college IT chess team for competition preparation.

2. NeuroTRIBE: Brain Activity Analysis for ADHD Research **GitHub Link**

- Developed a neuroimaging research pipeline to investigate differences in brain responses to movie stimuli between participants with and without ADHD.
- Automated the analysis of fMRI data using pretrained brain-activity models to identify patterns across brain regions and networks.
- Applied statistical testing with confounding-factor adjustments and generated 3D brain visualizations and reports to support research analysis.

3. Low-Cost Cattle Monitoring with ESP32 and DBSCAN **GitHub Link**

- Designed a low-cost cattle monitoring system that reduces dependence on individual GPS trackers by estimating animal proximity through wireless signal strength.
- Led development of an ESP32-based tracking prototype with a GPS-equipped master node, wireless child nodes, and a live dashboard for monitoring herd movement.
- Applied DBSCAN clustering to identify grazing zones and movement patterns, with telemetry processed through Flask and MongoDB and displayed in a Next.js dashboard.

EDUCATION AND ACHIEVEMENTS

- Qualified GATE DA 2026 and graduated in IT Engineering with an 8.5 CGPA from Mumbai University.
- Published an automation suite for CA and audit firms, adopted by two CA firms and used by approximately 20 auditors.